



Serviços Telemáticos

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Cofinanciado por:



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The Internet: a “nuts and bolts” view

Billions of connected computing *devices*:

- *hosts* = end systems
- running *network apps* at Internet’s “edge”a

Packet switches: forward packets (chunks of data)

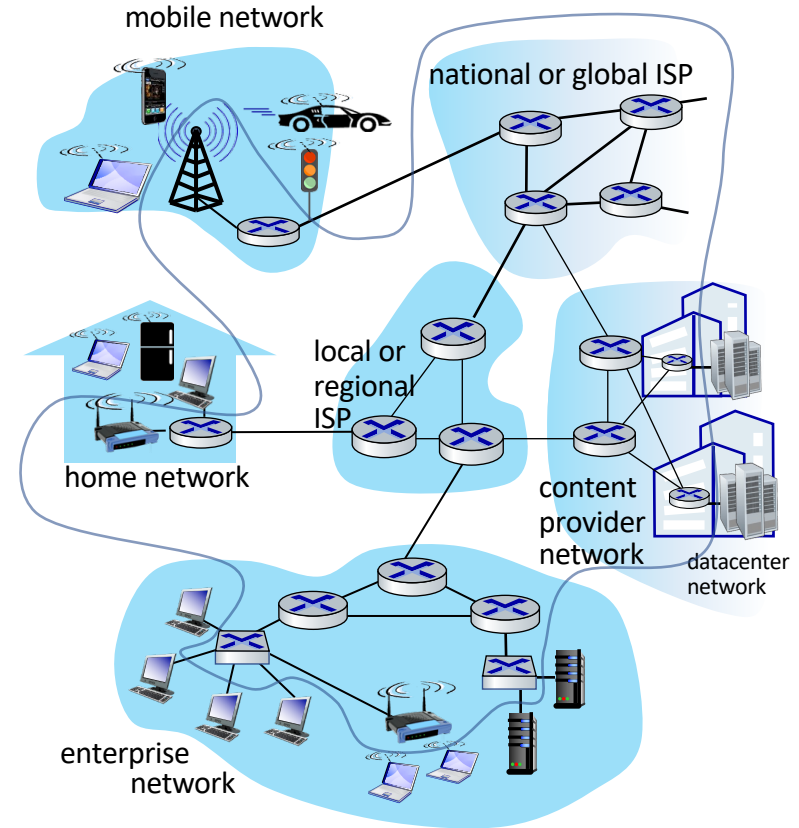
- *routers, switches*

Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

Networks

- collection of devices, routers, links: managed by an organization





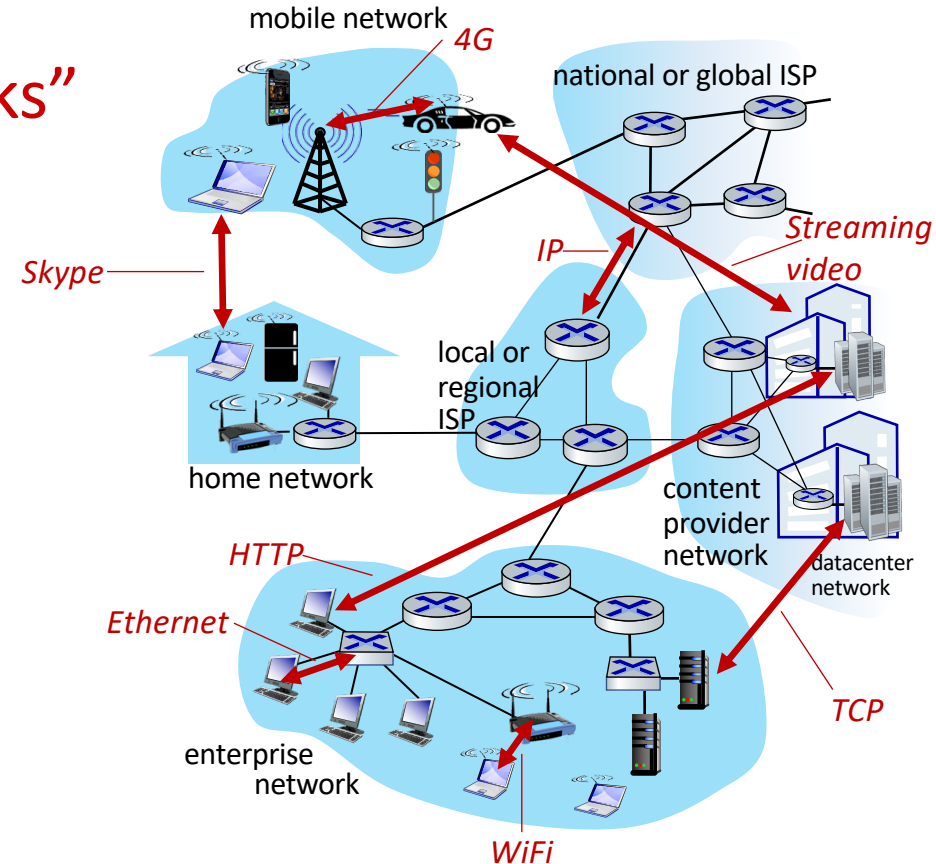
The Internet: a “nuts and bolts” view

- *Internet: “network of networks”*

- Interconnected ISPs
- *protocols* are everywhere
 - control sending, receiving of messages
 - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4/5G, Ethernet

- *Internet standards*

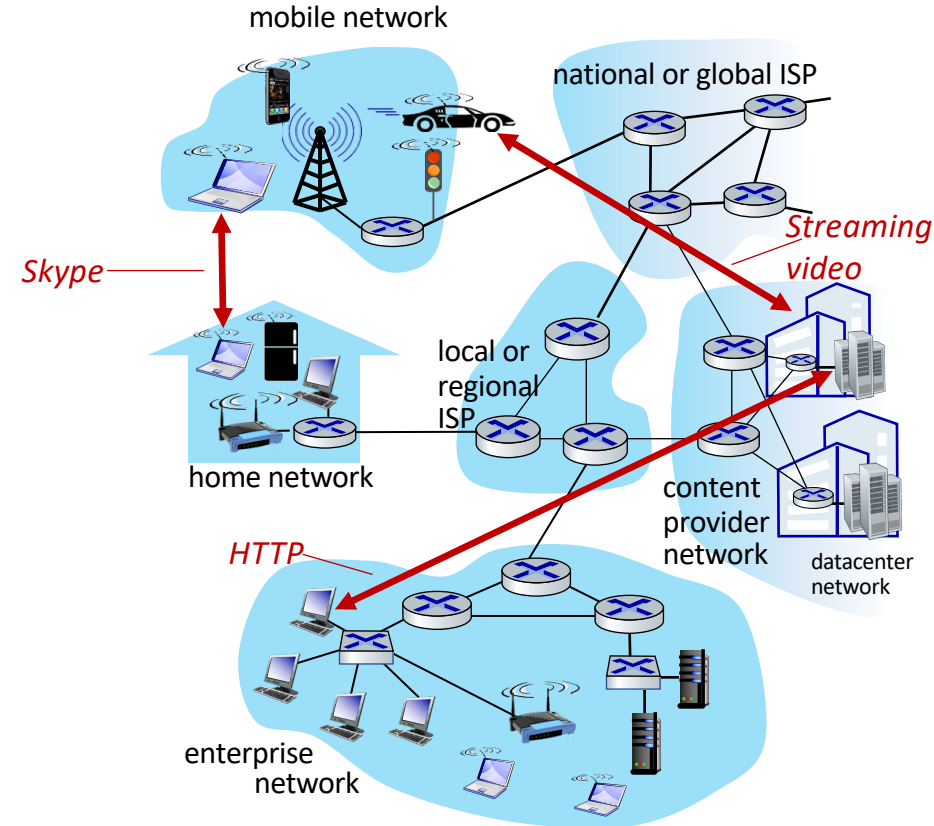
- RFC: Request for Comments
- IETF: Internet Engineering Task Force





The Internet: a “services” view

- *Infrastructure* that provides services to applications:
 - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...
- provides *programming interface* to distributed applications:
 - “hooks” allowing sending/receiving apps to “connect” to, use Internet transport service
 - provides service options, analogous to postal service





What's a protocol?

Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

Rules for:

- ... specific messages sent
- ... specific actions taken
when message received,
or other events

Network protocols:

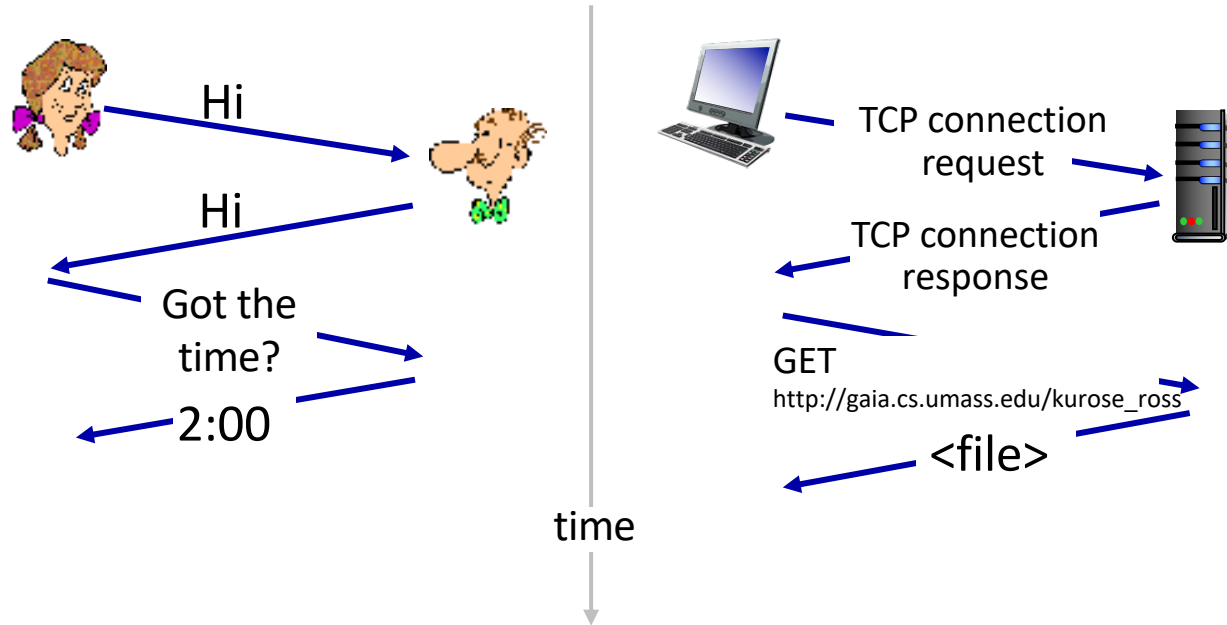
- computers (devices) rather than humans
- all communication activity in Internet
governed by protocols

*Protocols define the **format, order** of
messages sent and received among
network entities, and **actions taken**
on message transmission, receipt*



What's a protocol?

A human protocol and a computer network protocol:



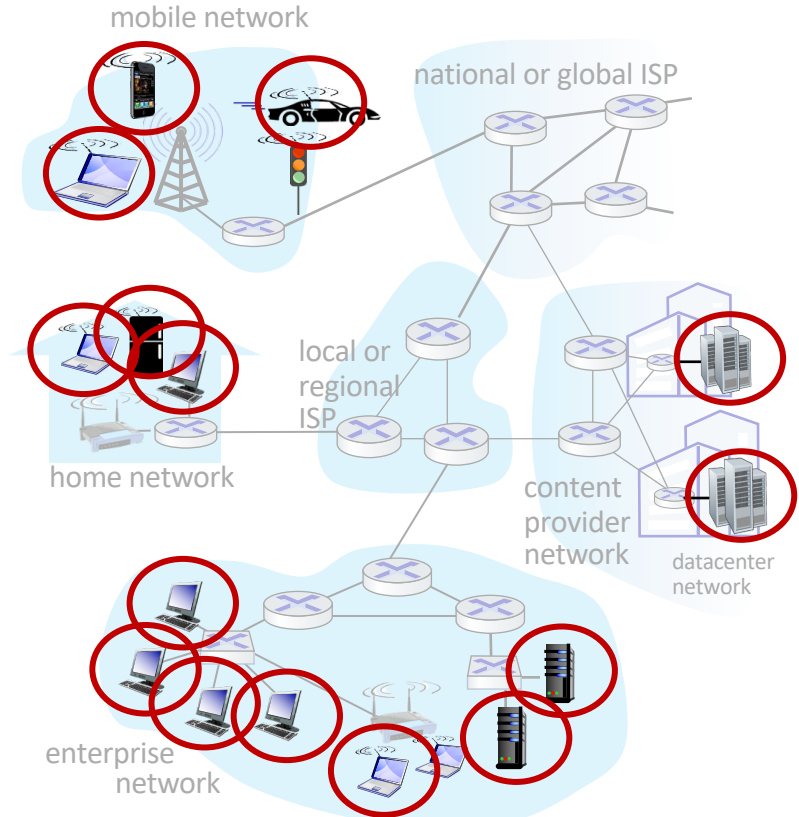
Q: other human protocols?



A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers





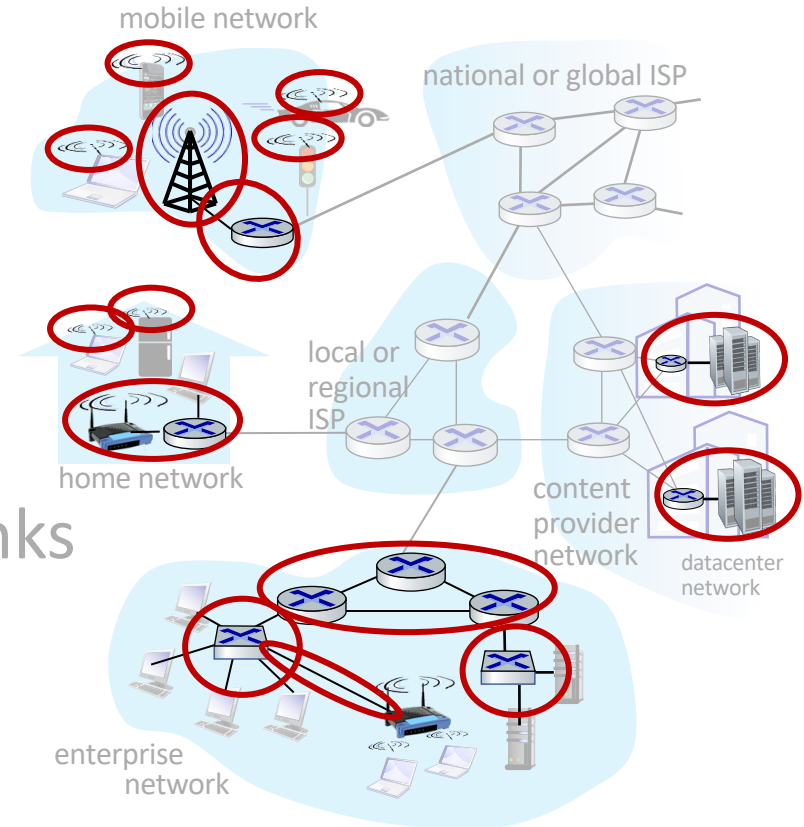
A closer look at Internet structure

Network edge:

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Access networks, physical media:

- wired, wireless communication links





A closer look at Internet structure

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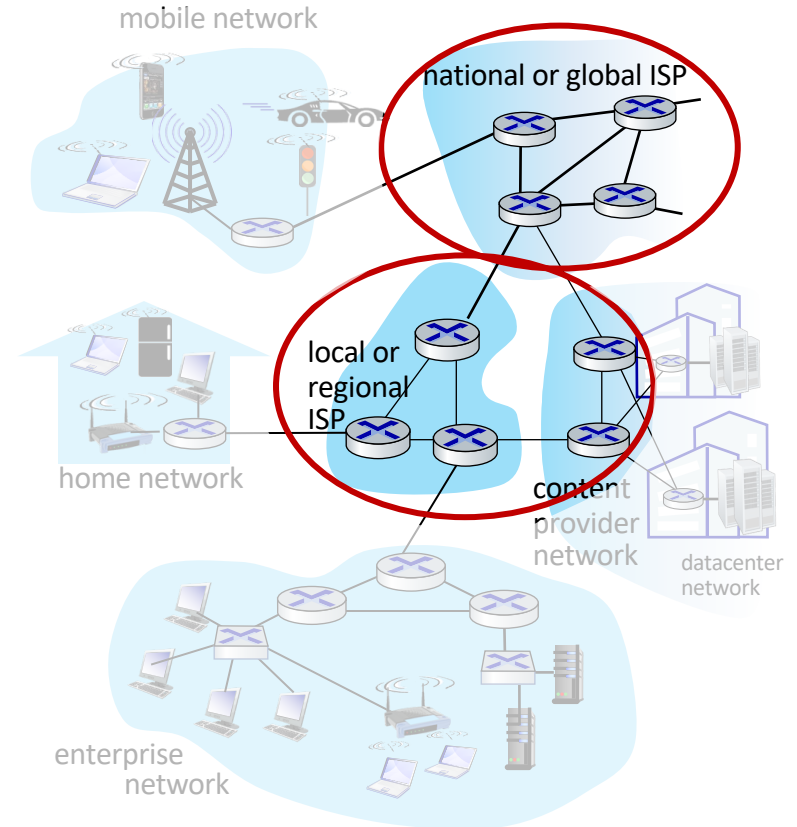
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Access networks, physical media:

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Network core:

- interconnected routers
- network of networks

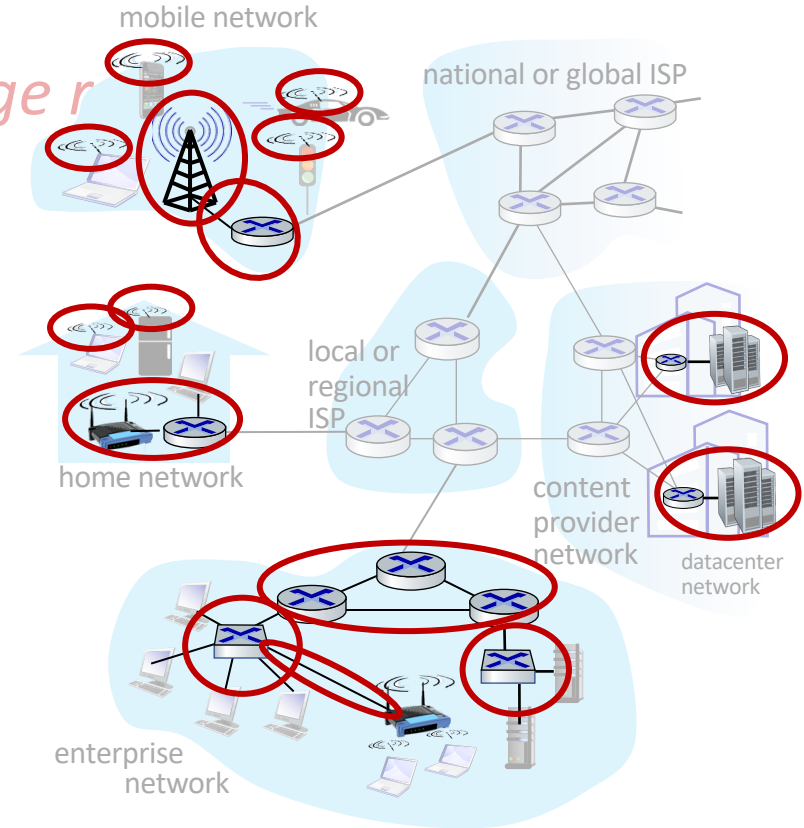




Access networks and physical media

Q: How to connect end systems to edge r

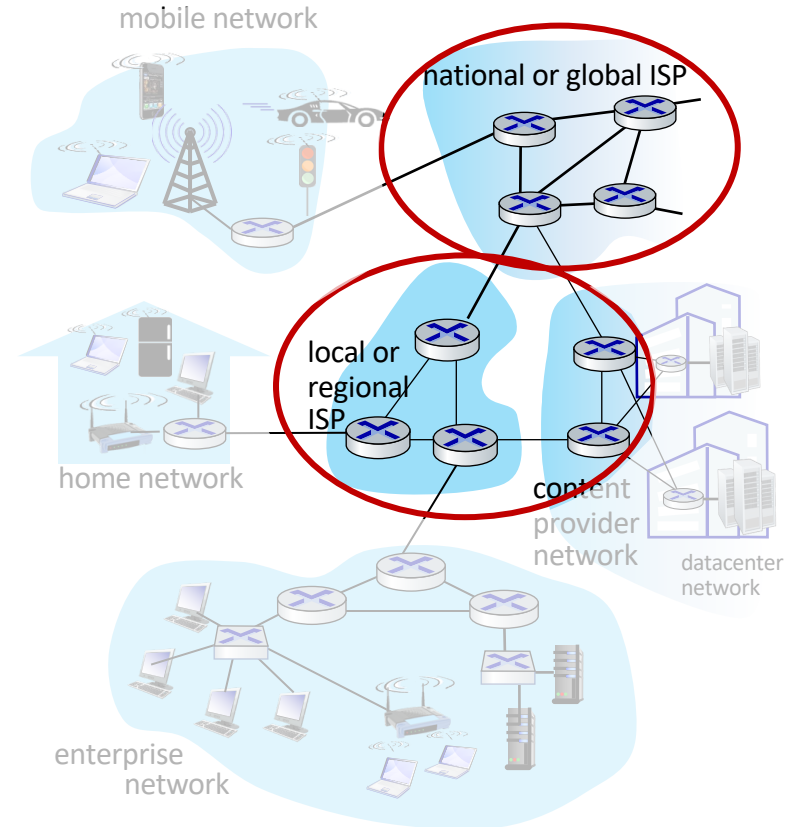
- residential access nets
- institutional access networks (school, company)
- mobile access networks (WiFi, 4G/5G)





The network core

- mesh of interconnected routers
- **packet-switching**: hosts break application-layer messages into *packets*
 - network **forwards** packets from one router to the next, across links on path from **source to destination**

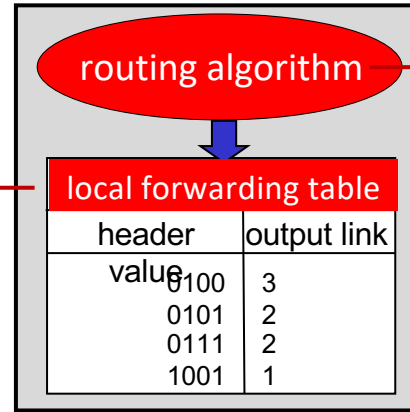




Two key network-core functions

Forwarding:

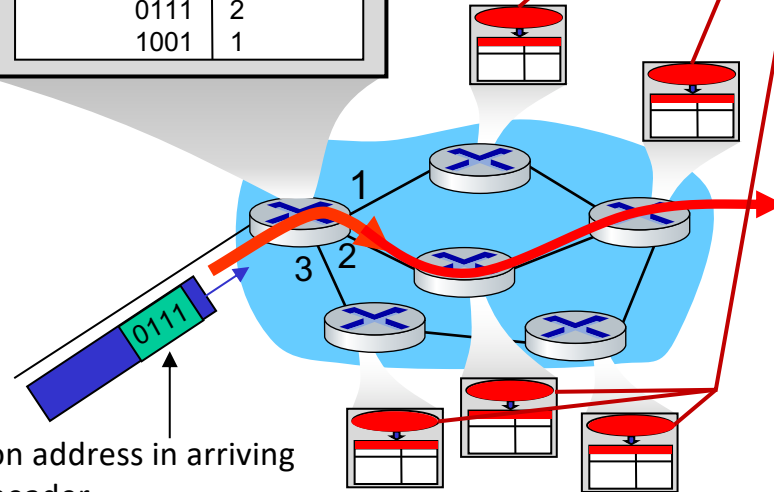
- aka “switching”
- *local* action: move arriving packets from router’s input link to appropriate router output link



destination address in arriving packet's header

Routing:

- *global* action: determine source-destination paths taken by packets
- routing algorithms



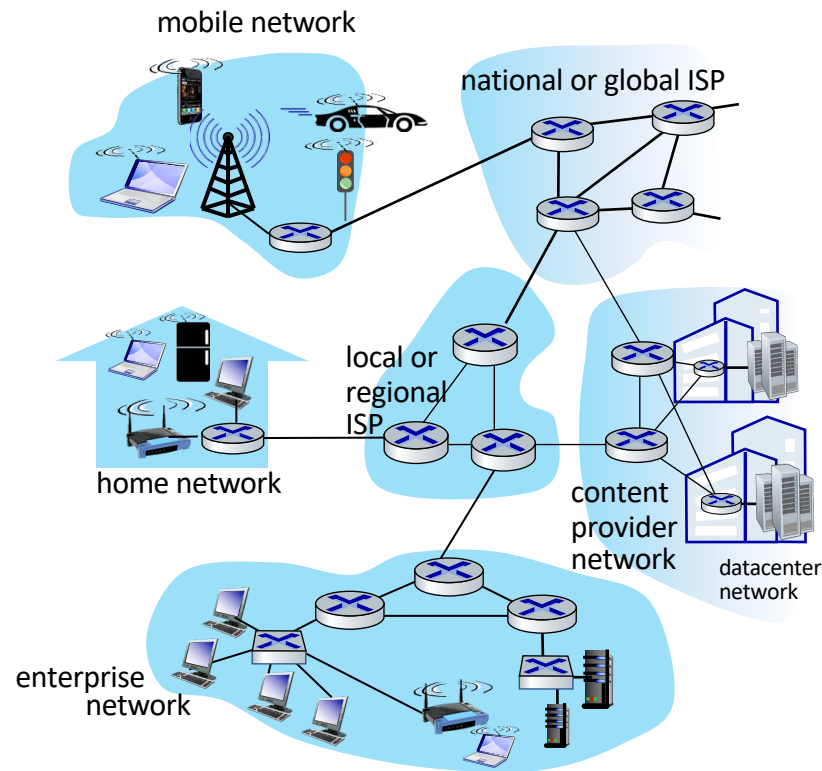




Internet structure: a “network of networks”



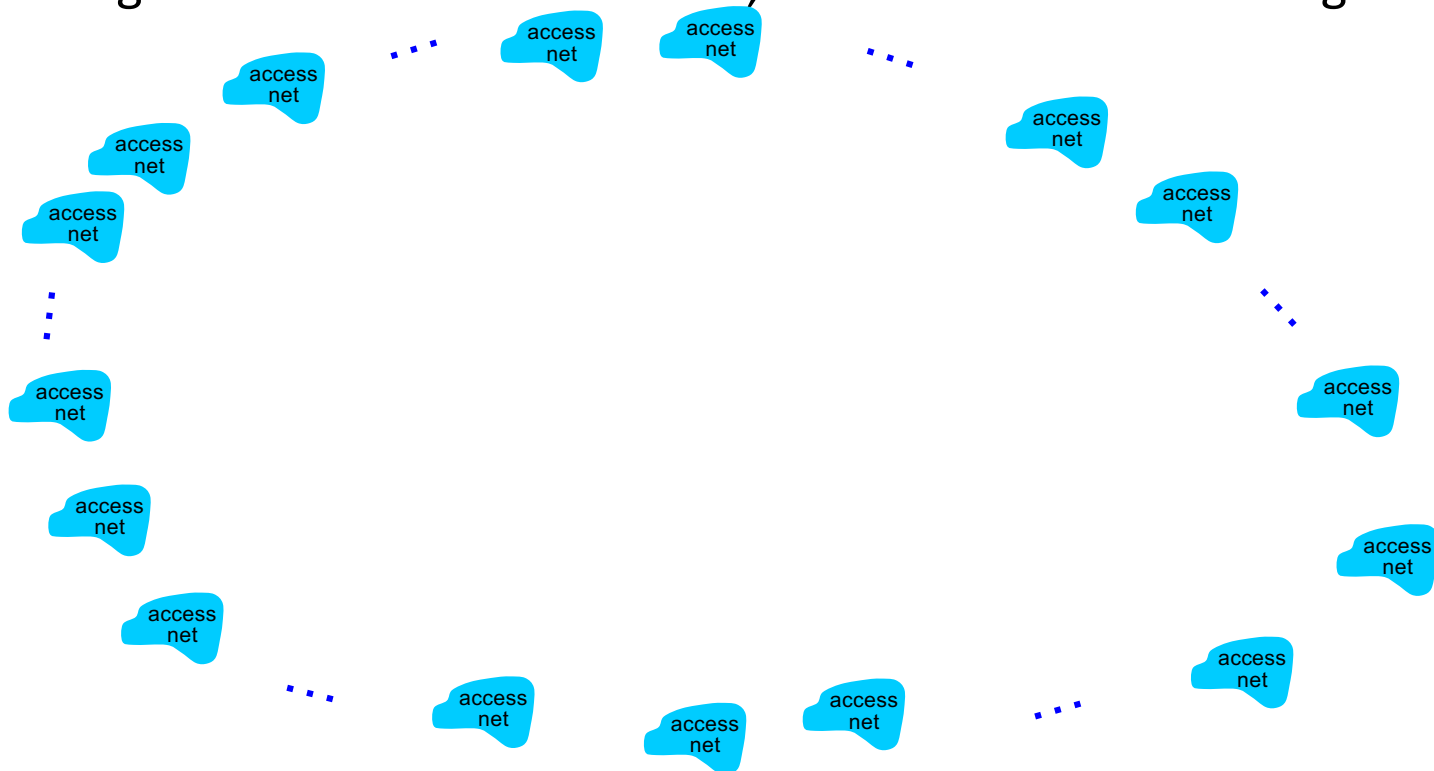
- hosts connect to Internet via **access** Internet Service Providers (ISPs)
- access ISPs in turn must be interconnected
 - so that *any* two hosts (*anywhere!*) can send packets to each other
- resulting network of networks is very complex
 - evolution driven by **economics, national policies**



Internet structure: a “network of networks”



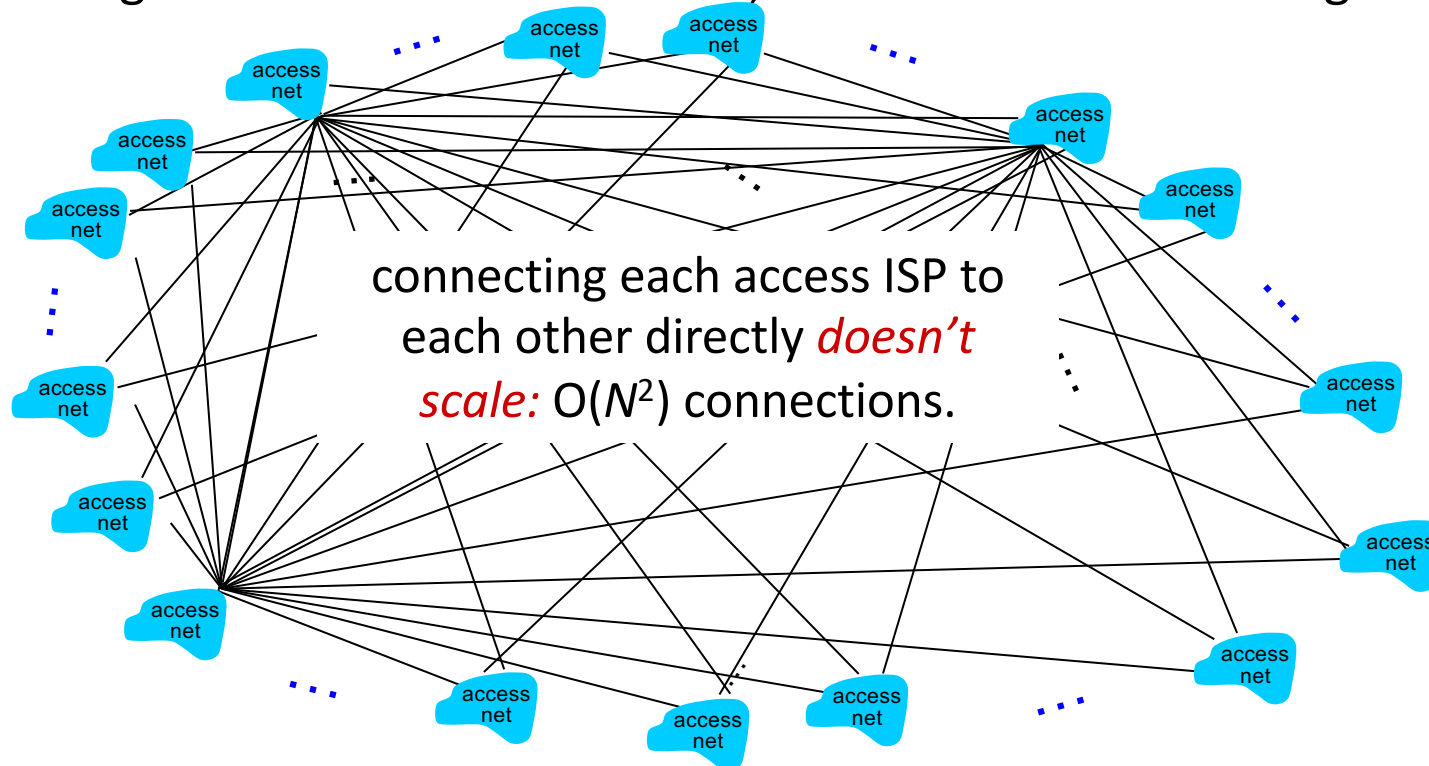
Question: given *millions* of access ISPs, how to connect them together?





Internet structure: a “network of networks”

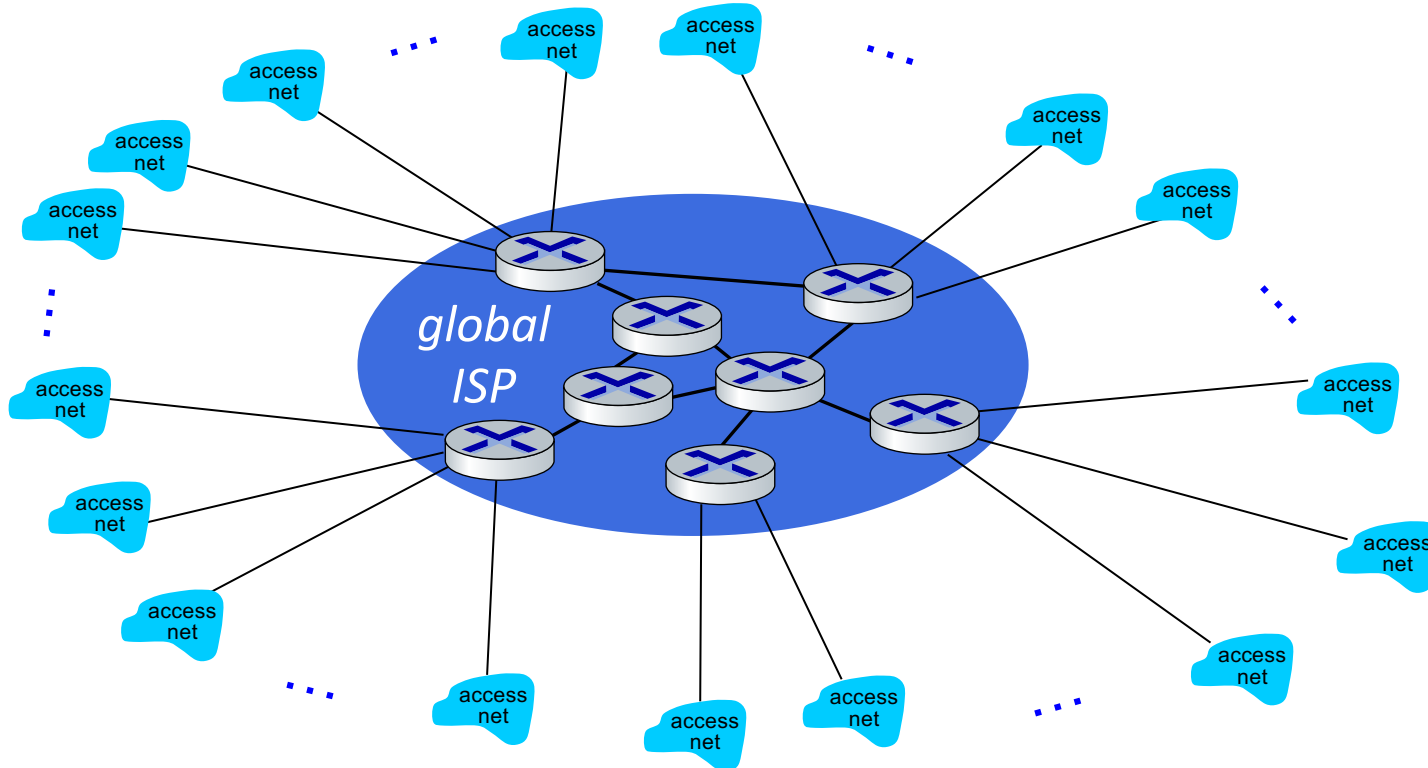
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Internet structure: a “network of networks”



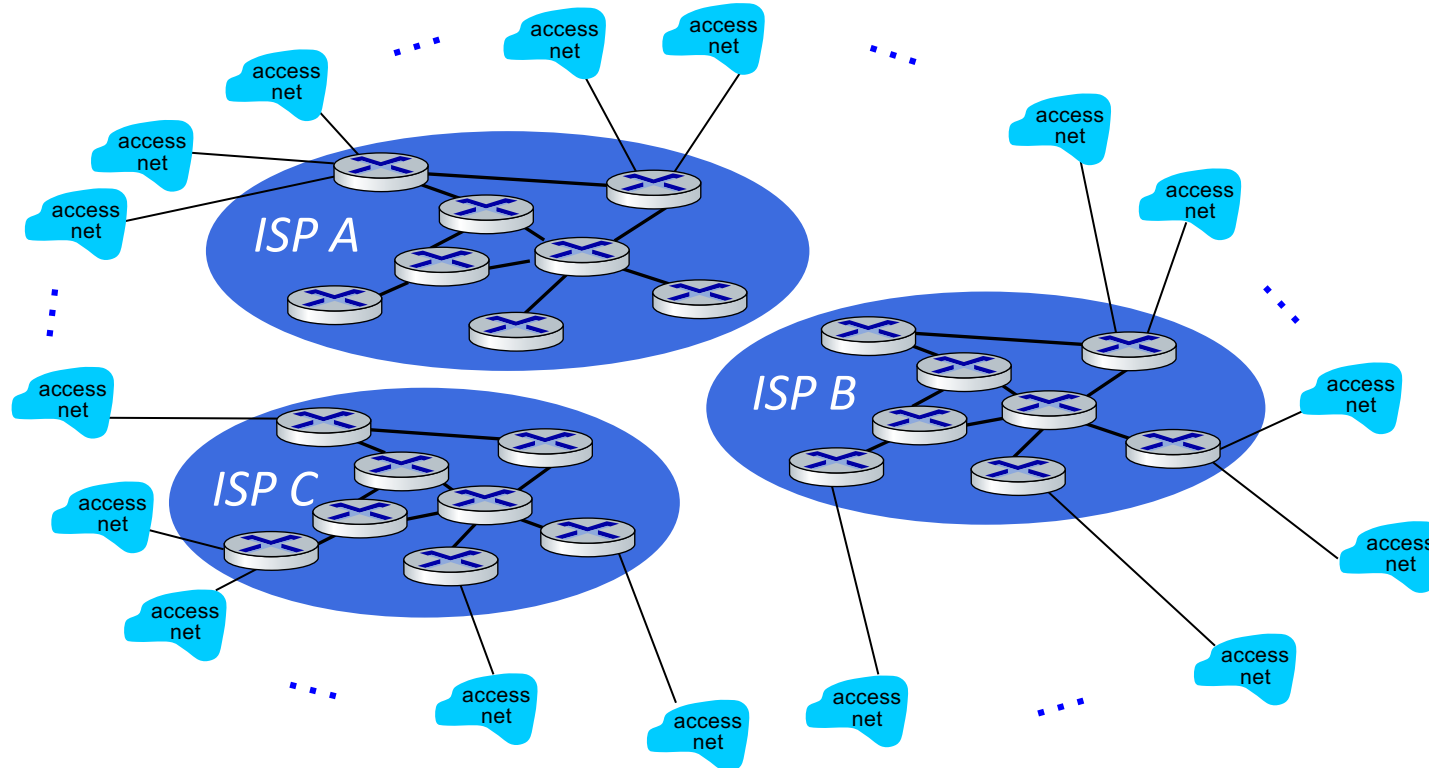
Option: connect each access ISP to one global transit ISP? Customer and provider ISPs have economic agreement.



Internet structure: a “network of networks”



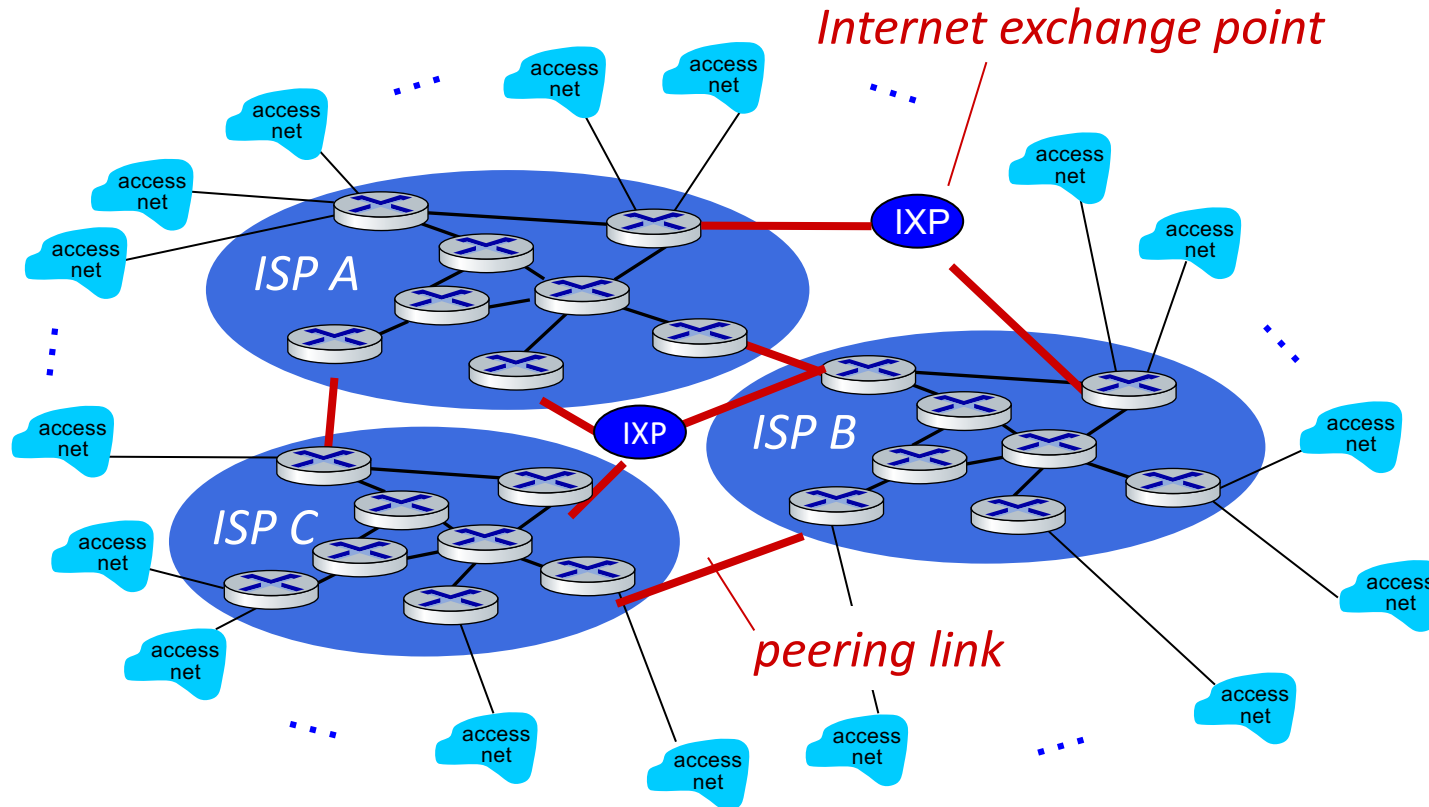
But if one global ISP is viable business, there will be competitors



Internet structure: a “network of networks”



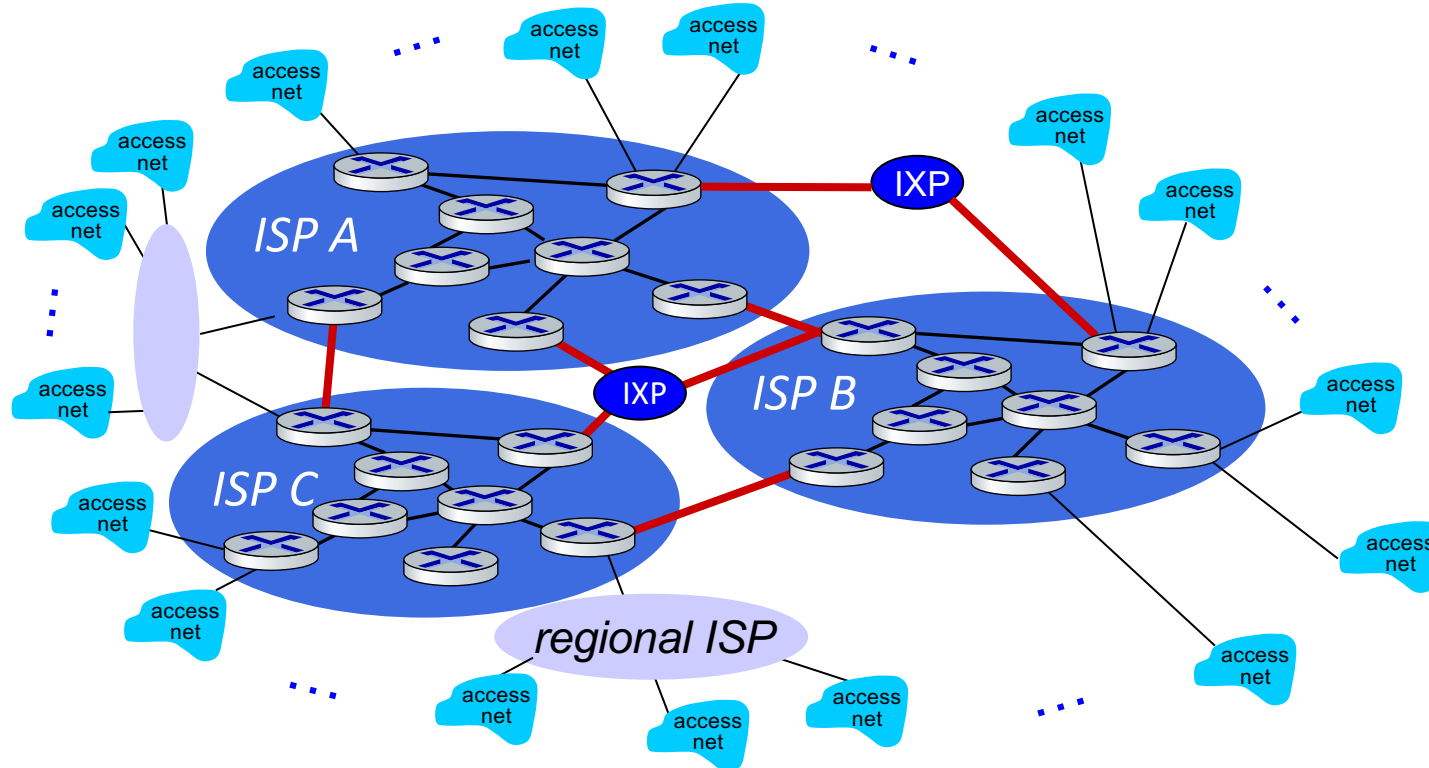
But if one global ISP is viable business, there will be competitors who will want to be connected



Internet structure: a “network of networks”



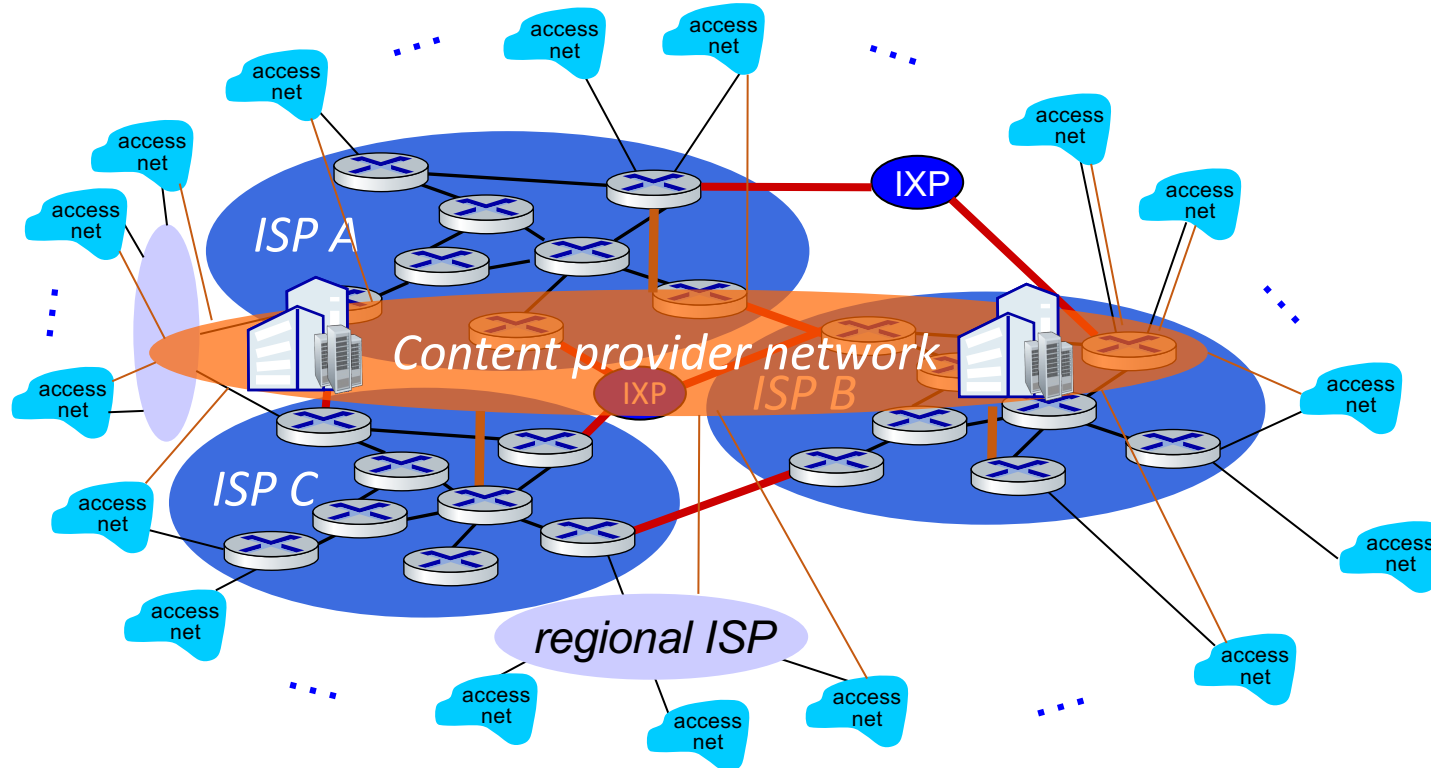
... and regional networks may arise to connect access nets to ISPs



Internet structure: a “network of networks”

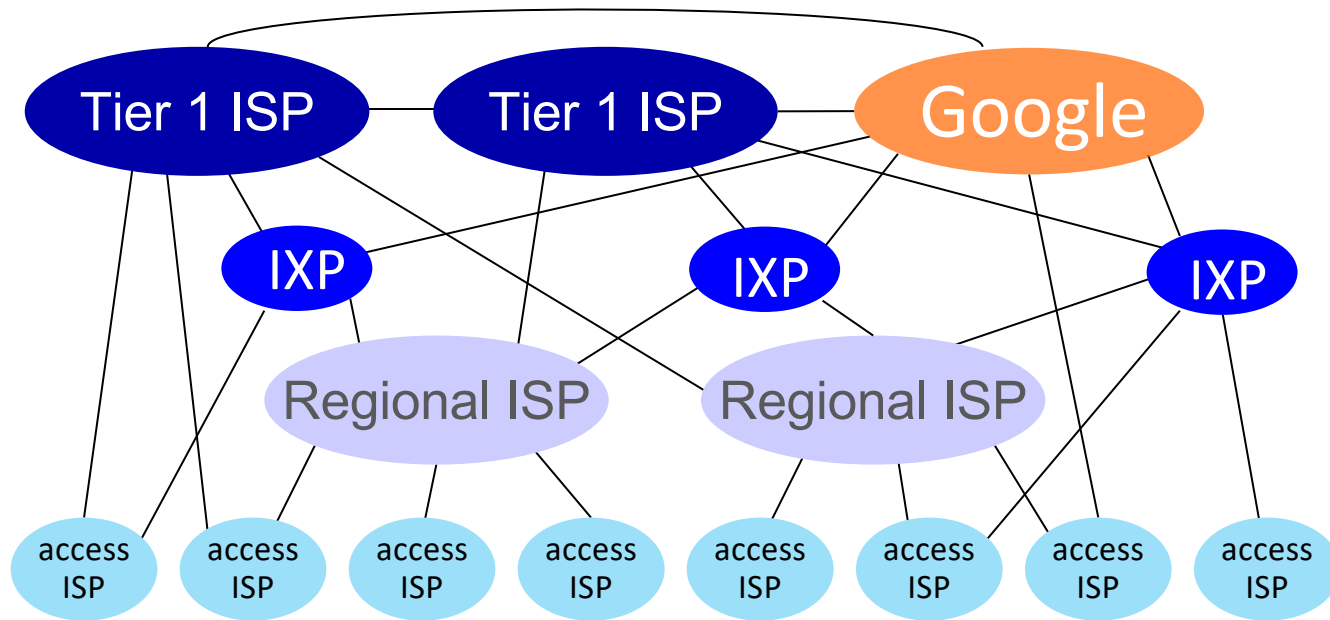


... and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users





Internet structure: a “network of networks”



At “center”: small # of well-connected large networks

- **“tier-1” commercial ISPs** (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- **content provider networks** (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs